Database Architecture

Mirunalini.P

SSNCE

December 23, 2025

Session Objective

- To learn about Data Base Architecture

Session Outcome

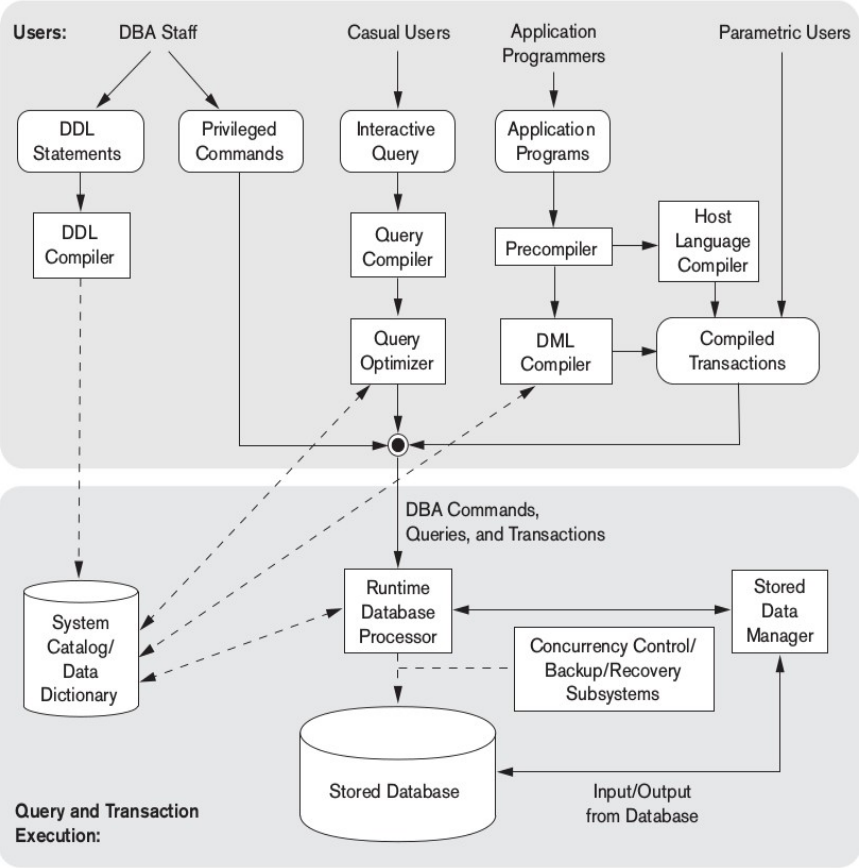
At the end of this session, participants will be able to:

- Understand the Data Base Architecture

Table of Contents

1. Database System Enivornment
2. DBMS Component Modules
3. Database Users
4. Centralized and Client/Server Architectures for DBMSs
5. Three Tier Architecture
6. Reference

Database Enivornment



DBMS Component Modules

- A DBMS is a complex software system, constitutes different components.
- The top part of the components consists of different users and their interfaces.
- The lower part shows the internal modules of the DBMS responsible for data storage and processing of transactions.

System Catalog

- The database and the DBMS catalog are usually stored on disk.
- Access to the disk is controlled primarily by the **operating system (OS)**, which schedules disk read/write.
- Many DBMSs have their own **buffer management module** to schedule disk read/write.
- Reducing disk read/write **improves performance** considerably.
- **Data manager module** of the DBMS controls access to DBMS information that is stored on disk, whether it is a part of the database or the catalog.

Database Users

Users are differentiated by the way they expect to interact with the system

- Database Administrators
  Coordinates all the activities of the database system which includes enterprise's information resources and needs.
  - Schema definition
  - Storage structure and access method definition
  - Schema and physical organization modification
  - Granting user authority to access the database
  - Periodical backup to prevent loss of data
  - Ensuring enough free disk space is available and upgrading disk space
  - Monitoring performance and responding to changes in requirements

DDL Compiler

DBA staff works on defining the database and tuning it using the DDL and other privileged commands.

- The DDL compiler processes schema definitions, specified in the DDL
- Stores descriptions of the schemas (meta-data) in the DBMS catalog.
- The catalog includes information such as:
  - Names and sizes of files
  - Names and data types of data items
  - Storage details of each file
  - Mapping information among schemas, and constraints.
- The catalog also stores many other types of information that are needed by the DBMS modules

Casual users

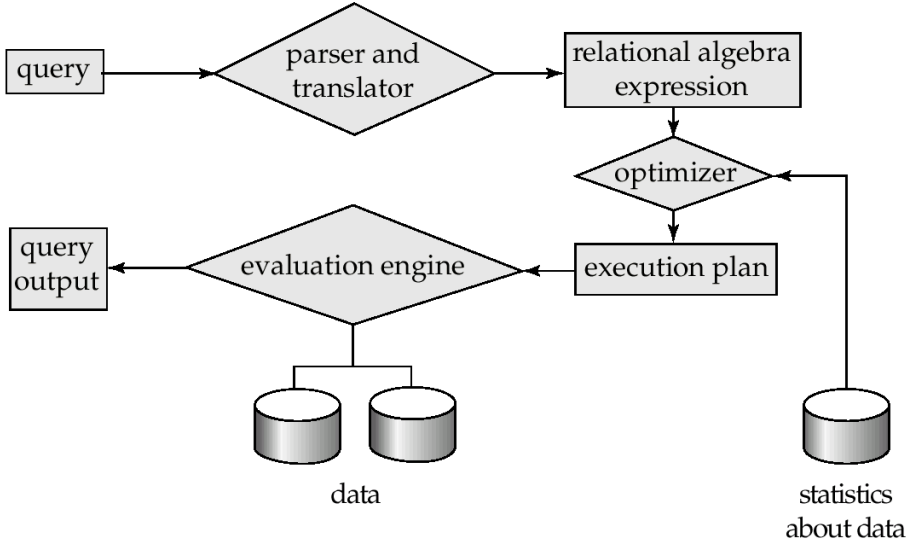
- People with an occasional need for information.
- Interact using the interactive query interface.
- Do data entry by supplying parameters for predefined transactions. (canned)

These queries are **parsed and validated for correctness** of the query syntax, the names of files and data elements

Query compiler compiles them into an **internal form**. The query optimizer is concerned with:

- Rearrangement and possible reordering of operations, elimination of redundancies, and use of correct algorithms and indexes during execution.
- It consults the system catalog for statistical and other physical information
- Generates executable code that performs the necessary operations for the query
- Makes calls on the runtime processor.

Query Processor



Application programmers Parametric users

- Interact with system through application programs.
- Application programmers write programs in host languages such as Java, C, or C++ that are submitted to a precompiler.
- The precompiler extracts DML commands from an application program written in a host programming language.
- Commands sent to DML compiler for compilation into object code.
- The rest of the program is sent to the host language compiler.
- The object codes for the DML commands and the rest of the program are linked, forming a canned transaction
- Executable code make calls to the runtime database processor.
- Canned transactions are executed repeatedly by parametric users via PCs or mobile apps; these users simply supply the parameters to the transactions.

Runtime database processor

The runtime database processor executes :

- The privileged commands
- The executable query plans
- The canned transactions with runtime parameters.
- It works with the system catalog and may update it with statistics.
- It also works with the stored data manager

Storage Manager

A program module that provides the interface between:

- the lowest level data stored in the database and the application programs and queries submitted to the system
- Translates the DML statements into low-level file-system commands

Storage manager components:

- **Authorization and Integrity Manager:** Tests integrity constraints and checks the authority of users to access data.
- **Transaction Manager:** Ensures that the database remains in a consistent state despite failures and concurrent transaction executions proceed without conflicting.
- **File Manager:** Manages the allocation of space on disk storage and the data structures used to represent information stored on disk.
- **Buffer Manager:** Responsible for fetching data from disk into main memory, deciding what data to cache in main memory.

Centralized Architectures for DBMSs

- All database processing and storage are handled by a single central server.
- Users access the database through terminals or client machines with minimal or no processing.
- Ensures strong data consistency and centralized control.
- Easy to manage, secure, and maintain since data is stored in one location.
- Suffers from single point of failure—if the server fails, the entire system is unavailable.
- Performance may degrade when many users access the system simultaneously.
- Commonly used in early database systems and small-scale applications.

Two Trier Architecture

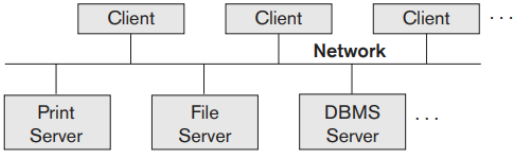


Figure 2.5
Logical two-tier client/server architecture.

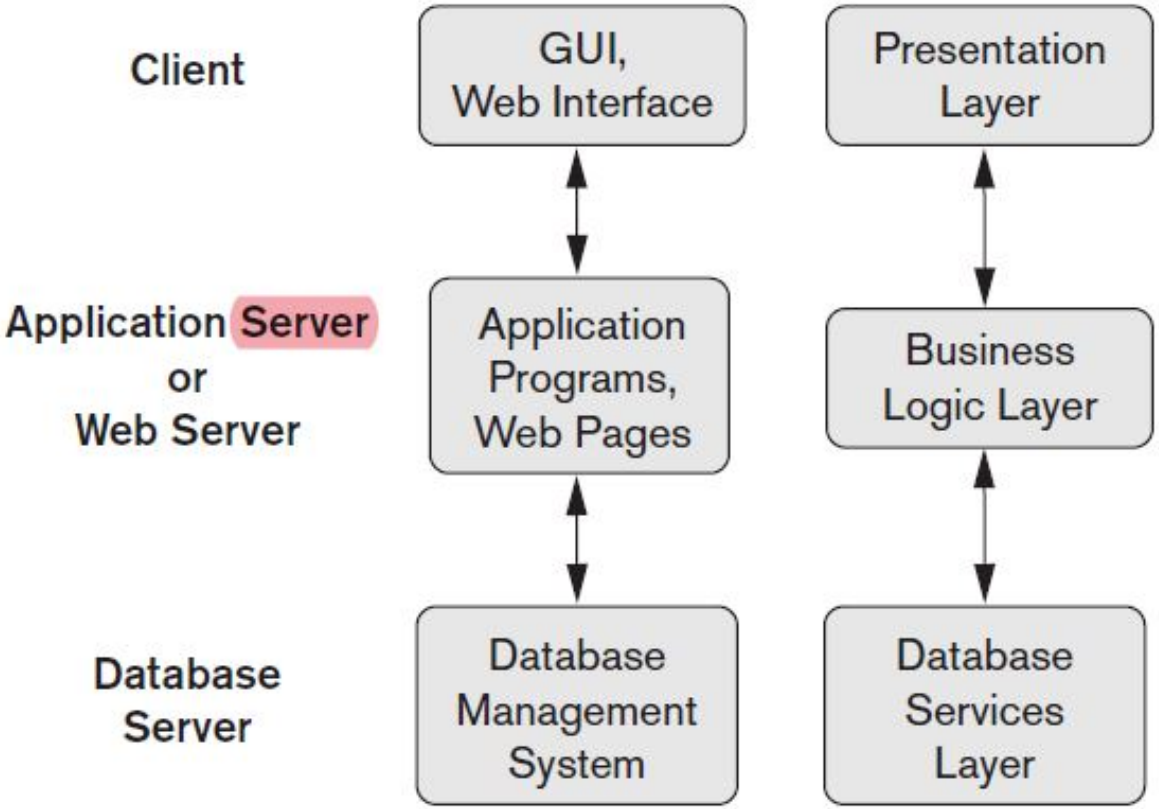
- It is common to have the client program that accesses the DBMS running on a separate computer or device from the computer on which the database resides.
- The former is called the client computer running DBMS **client software** and the latter is called the **database server**.

Two Trier Architecture

Client and Server Architecture:

- User interface programs and application programs can run on the client side.
- DBMS access is required establish a connection between client and server using ODBC-API.
- Application programming interface (API), which allows client-side programs to call the DBMS, as long as both client and server machines have the necessary software installed.
- The client sends requests which is process at the server side.
- Server can provide access to the client machines contains functions related to query and transaction processing.
- The architectures described here are called two-tier architectures because the software components are distributed over two systems: **client and server**

Three Trier Architecture



Three Tier Architecture

- Middle Tier:Middle tier plays an intermediate role by
  - Running application programs and business rules that are used to access data from the database server.
  - Checking client's credentials
  - Intermediate server accepts requests from the client, process the requests and sends database queries and commands to the server
  - Also act as a conduit by passing processed data to the user
- Client: Contains interfaces and webbrowsers

Reference

Fundamentals of Database systems 7th Edition by Ramez Elmasri.